Luis Fernando Villalon

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EDUCATION

San Diego State University *Expected May 2027*

Bachelor of Science in Computer Science, San Diego, CA

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Machine Learning, Data Science, Software Security, Data Privacy, UNIX System Administration

Activities & Organizations: ColorStack (SDSU Chapter Founder), SHPE, CodePath, SDx

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

Frameworks & Libraries: React, Next.js, Node.js, Express, FastAPI, Pydantic

Infrastructure & Tools: PostgreSQL, MongoDB, Supabase, Firebase, REST APIs, Git

EXPERIENCE

Software Engineer Intern *May 2024 - Aug 2024*

Ectron Corporation, San Diego, CA

- Built the frontend of a full-stack asset-tracking platform (React, Next.js, Node.js, PostgreSQL) that displays Azure IoT sensor data on an interactive Azure map.
- Integrated Azure services with the company's CMS after finding in documentation and forums that Next.js was the required path.
- Led two interns by defining their tasks and reviewing their implementation.

Computer Science Intern *Nov 2025 - Jan 2026*

Second Course, San Diego, CA (Hybrid)

- Reorganized and cleaned up a React Native app's codebase (TypeScript, Firebase) by removing unused files and simplifying its structure.
- Proposed food-information and directions features for an MVP serving 300+ college students; the founder accepted them. The app is live on the Apple App Store.

Front-End Developer Intern *Dec 2024 - Mar 2025*

Sports Excitement LLC, New York, NY (Remote)

- Built React and Tailwind CSS interfaces from Figma designs, including the login page and key form components, for a site that is live.
- Reported progress and picked up next tasks in a 12-plus-person Agile team.

Undergraduate Researcher *Jan 2026 - May 2026*

AI4Business Lab, San Diego, CA (Remote)

- Conducted a literature review on trust, safety, and adversarial robustness in AI agents, and summarized findings that informed a three-layer trust framework in an unpublished draft paper.

LEADERSHIP

Outreach Chair *Feb 2026 - Present*

ColorStack (SDSU Chapter), San Diego, CA

- Lead recruitment and digital outreach across social media, email, and Discord, connecting members with industry career opportunities.

PROJECTS

ROSS | *Node.js, SQLite, RAG, BM25, Clio API* ***1st Place**, Swans Applied AI Hackathon 2026*

- Built a RAG pipeline over case files: BM25 retrieval over extracted PDF text, with an Ask endpoint that returns "Not in the file" when the documents do not answer the question.
- Built citation verification so every claim links to its source passage; with an API key, the model's answer is checked to cite only passages it was given. Built a timeline view and simplified dashboard that cut case review from an afternoon to an estimated 90-second session; 1st of over 60 teams.

kanso | *Next.js, React, TypeScript, FastAPI, PostgreSQL* *Live app: kanso-landing.vercel.app*

- Built solo. Frontend uses separate data contexts for tasks, tags, habits, notes, and events so updates do not re-render unrelated views; failed saves roll back optimistic updates.
- Backend (FastAPI, SQLAlchemy, Alembic) verifies Supabase JWTs against JWKS and scopes every query by user_id, with per-user rate limits and AES-256-GCM encryption for third-party tokens.
- Separate AI service for learning recommendations, with links checked so the model does not invent URLs.

Agentic Abstraction for Chip Engineering (AACE) | *LangGraph, FastAPI, Next.js* *NXP Bootcamp, Agentic AI in the Cloud*

- Built the RTL-to-JSON and verification agents in a five-person team's multi-agent pipeline that turns SystemVerilog into validated JSON and architecture diagrams; failed checks loop back to the Architect for up to 3 attempts.
- Built a FastAPI backend with RTL upload and regeneration endpoints that reuse verified JSON, rendering SVG through the QuickChart Graphviz API.
- Used AWS Bedrock as the default LLM backend and EC2 and S3 for hosting and storage; Gemini and OpenAI are selectable through a config setting.